ALLAN SURAJ A

Bengaluru, India | +91 74830 84802 | allansuraj.4821@gmail.com | github.com/furious0420 | linkedin.com/in/allansuraj

Education

PES University Bengaluru, India
B.Tech in Electronics and Communication Engineering | CGPA: 8.08 / 10 *2023 – 2027 (Expected)*
- **Relevant Coursework:** Computer Architecture, Digital System Design, Machine Learning, DBMS

Technical Skills

Languages: C, C++, Python, SystemVerilog/Verilog, SQL, MATLAB
Hardware & Systems: RTL design, Xilinx Zynq SoC, Vivado, FPGA prototyping, Cadence Virtuoso
AI / Machine Learning: CNNs (YOLOv8, ResNet50), computer vision, object detection, Grad-CAM
Software & Tools: Linux, Git, debugging & profiling, relational databases, React
Security: OS hardening, security-policy implementation, attack-surface analysis, configuration management

Projects

Hardware–Software Co-Design for CNN Acceleration | *Xilinx Zynq SoC, Winograd* *Jun 2025 – Present*
- Porting a CNN inference pipeline onto a Xilinx Zynq SoC, replacing direct convolution with a modified Winograd algorithm to cut 2D convolution latency on edge hardware.
- Wrote the Winograd transform unit and the processing-element (PE) array in Verilog RTL.
- Splitting the workload between the programmable logic and the on-chip ARM processor, and laying out the datapath and memory hierarchy to fit the available logic and BRAM.

4-Bit ALU in 65/45 nm CMOS | *Cadence Virtuoso*
- Designed a transistor-level 4-bit ALU, building each gate from scratch and verifying every opcode in simulation.

Uncertainty-Aware Medical Abnormality Detection & Grounding | *Python, Computer Vision*
- Built a pipeline that classifies and localises chest X-ray abnormalities directly from the image, without relying on a prior clinical report.
- Added uncertainty estimation so low-confidence cases go to a human reviewer, with Grad-CAM heatmaps showing what the model looked at.

EO/IR Smart Surveillance System | *Python, YOLOv8, ResNet50* *HAL Aerothon 2025*
- Built a threat-detection system across RGB, thermal and aerial imagery using four models, sending each frame to the model suited to that input instead of one general detector.

Experience

Bharat Electronics Limited (BEL) Bengaluru, India
Student Intern – Network Systems & Cybersecurity *Jul 2025 – Aug 2025*
- Audited Windows 10 systems against security baselines and applied group-policy controls in a defence setup.
- Worked with senior engineers on OS-level performance tuning and system administration on enterprise systems.

PESU Venture Labs Bengaluru, India
Research Intern – Deep-Tech Venture Research *Jun 2025 – Aug 2025*
- Researched semiconductor and deep-tech trends to frame problem statements with startup potential.
- Interviewed senior semiconductor industry experts to test technical and market assumptions, and wrote the findings into structured briefs.

Hackathons

Terrathon – FinSync | *React, Accessible UI* *Oct 2025*
- Built a React banking app for senior citizens with a high-contrast UI and Kannada support; payments above a set amount stay on hold until a trusted contact approves them.

Horcrux – SuperPowerGlasses | *Computer Vision, Embedded Systems* *Sep 2024*
- Prototyped assistive smart glasses that stream camera and mic input to a laptop and return real-time audio and visual prompts for users with autism or visual impairment.

Certifications

Semiconductor Manufacturing (IISc) • ML Specialization (Coursera) • Google Cybersecurity Professional